# Acoustic Voice-Quality Features for Speech Analysis

Jitter, Shimmer, HNR, CPP, and GNE

Technical Report

**Abstract**

This report describes five families of acoustic voice-quality measurements used in speech analysis: jitter, shimmer, harmonics-to-noise ratio (HNR), cepstral peak prominence (CPP), and glottal-to-noise excitation (GNE). It explains the signal-processing concepts, equations, software operations, parameter choices, expected outputs, and limitations of the Python/Praat-based feature-extraction pipeline. The discussion is framed around research that may use speech features for Parkinson's disease (PD) classification. These measures are acoustic descriptors, not standalone diagnostic tests.

## Contents

# 1 Introduction

Speech is produced by an interacting source–filter system. Vocal-fold vibration generates an excitation signal, and the vocal tract shapes its spectrum. Speech recordings therefore contain information about periodicity, amplitude stability, harmonic structure, noise-like components, and spectral organization. Voice-quality features attempt to quantify selected aspects of these properties.

The features discussed here are complementary but not independent. Jitter and shimmer describe cycle-to-cycle perturbations; HNR describes harmonicity relative to noise; CPP describes the prominence of a cepstral periodicity peak; and GNE examines relationships between amplitude envelopes across frequency bands after an excitation-estimation stage.

# 2 Jitter: cycle-period perturbation

## 2.1 Vocal-fold pulse times and periods

Let the estimated glottal-cycle pulse times be

$$t_1, t_2, \ldots, t_M.$$

The corresponding periods are

$$T_i = t_{i+1} - t_i, \qquad i = 1, \ldots, M-1.$$

A perfectly periodic signal would have constant periods. Real speech exhibits small variations, and jitter quantifies selected forms of this variation.

In the code, a Praat point process is created using:

```
point_process = parselmouth.praat.call(
    sound, "To PointProcess (periodic, cc)", 60, 500
)
```

The command estimates pulse locations using a cross-correlation-based periodicity procedure. The values 60 and 500 specify the minimum and maximum expected fundamental-frequency range in hertz. They are analysis assumptions, not universal values for every speaker. If the pitch range is inappropriate, pulse detection can fail or produce octave errors, which then affect all pulse-based measures.

## 2.2 Local jitter

Local jitter is the mean absolute difference between consecutive accepted periods, normalized by the mean period:

$$J_{\text{local}} = \frac{\frac{1}{N-1}\sum_{i=1}^{N-1} |T_{i+1} - T_i|}{\overline{T}}.$$

It is dimensionless and is often reported as a ratio or percentage. It measures relative cycle-timing irregularity. Larger values indicate greater period perturbation, but do not identify its cause.

## 2.3 Absolute jitter

Absolute jitter is the mean absolute difference between adjacent periods:

$$J_{\text{abs}} = \frac{1}{N-1}\sum_{i=1}^{N-1} |T_{i+1} - T_i|.$$

It is measured in seconds. Unlike local jitter, it is not normalized by the average period. Consequently, the two measures are not directly interchangeable when comparing speakers with different fundamental frequencies.

### 2.4 RAP, PPQ5, and DDP

Relative Average Perturbation (RAP) compares each period with the average of a three-period neighborhood. A conceptual expression is

$$J_{\mathrm{RAP}} = \frac{1}{\overline{T}} \operatorname{mean}_i \left| T_i - \frac{T_{i-1} + T_i + T_{i+1}}{3} \right|.$$

PPQ5 uses a five-period moving average instead of a three-period average. These measures compare a period with its local neighborhood rather than only comparing adjacent periods.

DDP (difference of differences of periods) is related to RAP. Under the standard Praat definitions and identical accepted periods, DDP is three times RAP:

$$J_{\mathrm{DDP}} = 3J_{\mathrm{RAP}}.$$

Thus RAP and DDP are mathematically redundant up to scaling under matching conditions. Including both in a predictive model may add little independent information.

### 2.5 Praat command arguments

The jitter calls use the pattern:

```
value = parselmouth.praat.call(
    point_process, "Get jitter (local)",
    0, 0, 0.0001, 0.02, 1.3
)
```

The two zeros request the full object time interval. The minimum and maximum period arguments are 0.1 ms and 20 ms. The final parameter is the maximum permitted relative period factor used by Praat's acceptance logic. These settings affect which periods are retained. They should be held constant across recordings unless there is a documented reason to change them.

## 3 Shimmer: cycle-amplitude perturbation

### 3.1 Amplitude estimates

Let $A_i$ denote the amplitude assigned to the $i$th accepted vocal cycle. Shimmer measures changes in these amplitudes. Unlike jitter, shimmer requires both the waveform and the pulse locations because amplitude information must be extracted from the signal around each cycle.

### 3.2 Local shimmer

A conceptual expression is

$$S_{\mathrm{local}} = \frac{\operatorname{mean}_i |A_{i+1} - A_i|}{\overline{A}}.$$

It is a relative amplitude-perturbation measure. A larger value indicates less consistent cycle amplitudes.

### 3.3 Local shimmer in dB

The local-dB measure uses logarithmic amplitude ratios. A pairwise amplitude difference can be expressed as

$$\Delta A_{\mathrm{dB}} = 20 \log_{10} \left( \frac{A_{i+1}}{A_i} \right).$$

Praat's command averages the relevant absolute logarithmic differences according to its definition. The result is in decibels and should not be compared numerically with the dimensionless local-shimmer ratio.

### 3.4 APQ3, APQ5, APQ11, and DDA

Amplitude Perturbation Quotient (APQ) measures deviations from a moving average of cycle amplitudes. APQ3 uses three cycles, APQ5 uses five, and APQ11 uses eleven. A generic expression is

$$S_{\mathrm{APQ}K} = \frac{1}{\bar{A}} \operatorname{mean}_i \left| A_i - \frac{1}{K} \sum_{j \in \mathcal{N}_K(i)} A_j \right|,$$

where $\mathcal{N}_K(i)$ is the selected $K$-cycle neighborhood. DDA is related to APQ3; under standard matching definitions,

$$S_{\mathrm{DDA}} = 3S_{\mathrm{APQ3}}.$$

APQ3 and DDA are therefore related by scaling, not independent measurements.

The shimmer commands use the waveform and point process together:

```
local_shimmer = parselmouth.praat.call(
    [sound, point_process], "Get shimmer (local)",
    0, 0, 0.0001, 0.02, 1.3, 1.6
)
```

The final 1.6 parameter is the maximum permitted amplitude factor in the acceptance criteria. It is not a multiplier applied to the returned shimmer value.

## 4 HNR: harmonics-to-noise ratio

### 4.1 Concept

Voiced speech often contains a periodic component with harmonics at multiples of the fundamental frequency, together with noise-like components. HNR expresses the relative strength of harmonic and noise components:

$$\mathrm{HNR} = 10 \log_{10} \left( \frac{E_{\mathrm{harmonic}}}{E_{\mathrm{noise}}} \right).$$

Higher HNR generally indicates a more harmonic signal relative to its noise-like component. The equation gives the conceptual interpretation; Praat estimates harmonicity using its own analysis procedure.

### 4.2 Creating a harmonicity contour

```
harmonicity = call(
    sound, "To Harmonicity (cc)",
    0.01, 60.0, 0.1, 1.0
)
```

The arguments specify a 10 ms time step, a minimum pitch parameter of 60 Hz, a silence threshold of 0.1, and one period per analysis window. The command returns a time-varying Harmonicity object rather than a single scalar.

The recording interval is obtained from `sound.xmin` and `sound.xmax`. Four summaries are then calculated:

- Mean HNR: average harmonicity over the interval.
- Standard deviation: temporal variability of the HNR contour.
- Minimum HNR: lowest estimated harmonicity, using parabolic interpolation.
- Maximum HNR: highest estimated harmonicity, using parabolic interpolation.

The standard deviation describes variability in HNR, not variability in pitch. Minima and maxima may be sensitive to isolated noisy or invalid frames.

# 5 CPP: cepstral peak prominence

## 5.1 Cepstral representation

The cepstrum is derived from a logarithmic spectral representation transformed back to a time-like domain called quefrency. For voiced speech, regularly spaced harmonics can produce a cepstral peak near the fundamental period:

$$q_0 \approx \frac{1}{F_0}.$$

For example, $F_0 = 100$ Hz corresponds to a period of 10 ms. CPP measures how prominent a selected cepstral peak is relative to a fitted background trend.

## 5.2 Power-cepstrum procedure

The implementation follows these steps:

```
spectrum = call(sound, "To Spectrum", "yes")
cepstrum = call(spectrum, "To PowerCepstrum")
cpp = call(
    cepstrum, "Get peak prominence",
    60.0, 333.3, "Parabolic",
    0.001, 0.05, "Straight", "Robust"
)
```

The spectrum is calculated first, then converted to a power cepstrum. The peak search range of 60–333.3 Hz corresponds approximately to periods from 16.7 ms to 3 ms. Parabolic interpolation estimates the peak location/value. The trend is fitted over the specified quefrency range using a straight-line model and robust fitting.

The supplied command returns one overall CPP estimate for the recording. Naming this value `mean_cpp` can be misleading because it is not an average of frame-wise CPP values. A name such as `overall_cpp` is more precise. A frame-wise CPP standard deviation requires calculating CPP separately over frames and then taking the standard deviation of those values.

# 6 GNE: glottal-to-noise excitation

## 6.1 Conceptual purpose

GNE is a distinct voice-quality measure. It examines relationships between amplitude envelopes in different frequency bands of an estimated excitation signal. It is not interchangeable with HNR or CPP. A typical conceptual pipeline includes inverse filtering, band-limited envelope extraction, and correlation analysis across separated frequency bands.

## 6.2 LPC inverse filtering

A simplified source–filter model is

$$X(z) = G(z)E(z),$$

where $X(z)$ is the speech signal, $G(z)$ is the vocal-tract transfer function, and $E(z)$ is the excitation. An inverse filter attempts to estimate

$$E(z) \approx \frac{X(z)}{G(z)}.$$

Linear predictive coding models a sample from previous samples:

$$x[n] \approx \sum_{k=1}^{p} a_k x[n-k].$$

The coefficients can be estimated from autocorrelation equations of the form

$$\mathbf{Ra} = \mathbf{r}.$$

The residual after inverse filtering is treated as an estimate of the excitation. It is not a direct measurement of the glottal source.

### 6.3 Hilbert envelopes and cross-correlation

For a band-limited signal $u(t)$, the analytic signal is

$$z(t) = u(t) + j\mathcal{H}\{u(t)\},$$

where $\mathcal{H}$ denotes the Hilbert transform. Its amplitude envelope is

$$A(t) = |z(t)| = \sqrt{u(t)^2 + \mathcal{H}\{u(t)\}^2}.$$

Comparing envelopes rather than raw waveforms focuses on slower amplitude fluctuations and reduces sensitivity to the differing carrier frequencies and phases of separate bands. Cross-correlation between two envelopes can be written conceptually as

$$R_{uv}[k] = \sum_{n} A_u[n] A_v[n-k].$$

A GNE implementation uses a prescribed set of bands, separation criteria, correlation normalization, lag handling, and aggregation procedure. These details are algorithm-specific and must be reproduced accurately.

### 6.4 Important implementation validation

The proposed Python fragment uses 10 kHz resampling, 30 ms frames, 10 ms hops, and LPC order 13. It then constructs correlation matrices, solves LPC systems, applies an inverse filter, computes band envelopes, and aggregates correlations. However, the exact tensor shapes and lag conventions returned by the selected SpeechBrain functions must be checked. The matrix construction, filter coefficient layout, dimensions passed to `lfilter`, and correlation-to-decibel transformation must match the reference algorithm. A line such as

$$-10\log_{10}(1-r)$$

is a particular transformation of a correlation value; it must not be assumed to be the standard GNE formula without verifying the source implementation. A threshold such as 10 dB also requires justification from the reference method.

Therefore, the supplied GNE fragment should be regarded as an implementation outline until its outputs have been checked against a validated implementation and known recordings. Do not report its values as verified GNE measurements solely because the code executes.

# 7 Custom jitter-to-shimmer ratio

A proposed derived feature is

$$R_{JS} = \frac{J_{\text{local}}}{S_{\text{local}}}.$$

It compares the numerical magnitudes of relative period and amplitude perturbations. It is not a standard Praat feature or an established clinical biomarker. The ratio should only be calculated when both values are finite and shimmer is positive. It can become unstable when shimmer is close to zero. Its usefulness must be tested empirically, including an ablation comparison with and without the ratio.

# 8 Feature inventory and dependencies

| Feature family | Interpretation and dependency |
|---|---|
| Local/absolute jitter | Adjacent cycle-period differences; requires reliable pulse detection. |
| RAP/PPQ5/DDP | Period perturbation relative to local neighborhoods; DDP is scaled RAP under standard definitions. |
| Local/local-dB shimmer | Adjacent cycle-amplitude variation; requires waveform and pulse process. |
| APQ3/APQ5/APQ11/DDA | Amplitude deviations from moving averages; DDA is scaled APQ3 under standard definitions. |
| HNR statistics | Summaries of a frame-wise harmonicity contour. |
| CPP | Prominence of a cepstral peak relative to a fitted trend; current implementation returns one recording-level estimate. |
| GNE | Cross-band envelope relationship after excitation estimation; requires a validated dedicated algorithm. |
| Jitter/shimmer ratio | Custom derived feature; may be unstable and redundant. |

# 9 Research considerations for PD speech classification

1. These are acoustic descriptors, not diagnostic tests. Their predictive value must be established using appropriate evaluation.

2. Use consistent sample-rate, pitch-range, period limits, amplitude thresholds, and preprocessing across recordings.

3. Inspect pulse tracks and harmonicity contours. Invalid pulse detection can affect several features at once.

4. Handle undefined measurements explicitly. `NaN` means unavailable, not zero. Imputation parameters must be learned from training data only.

5. Some features are mathematically related: RAP/DDP and APQ3/DDA are scaled pairs under matching definitions. Check collinearity and feature importance.

6. If multiple recordings belong to one speaker, use speaker-independent splits where possible to reduce identity leakage.

7. Recording gain, microphone distance, background noise, phonetic content, and task design can affect the measurements.

## 10 Conclusion

Jitter and shimmer quantify cycle-level timing and amplitude perturbations; HNR summarizes harmonicity relative to noise; CPP quantifies cepstral peak prominence; and GNE examines cross-band excitation-envelope relationships. Each captures a different aspect of voice production, but their reliability depends on the underlying signal analysis. In particular, GNE must be implemented and validated according to a precise reference algorithm before it is used as a research feature. Consistent preprocessing, careful handling of undefined values, and speaker-independent evaluation are essential for meaningful PD speech-classification experiments.

## References


[1] P. Boersma and D. Weenink, *Praat: doing phonetics by computer*, software and documentation, https://www.fon.hum.uva.nl/praat/.

[2] Y. Jadoul, B. Thompson, and B. de Boer, ‘Introducing Parselmouth: A Python interface to Praat," \emphJournal of Phonetics, vol. 71, pp. 1-15, 2018. \bibitemspeechbrain SpeechBrain, ‘Voice analysis and preprocessing tutorial,” documentation, https://speechbrain.readthedocs.io/.